

Analysis of Sequence Variation

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Whole Genome/Exome Sequencing

Whole Genome Sequencing

- ▶ **W**hole **G**enome **S**equencing (WGS)
- ▶ Generate enough reads to attain:
 - ▶ More than 95% coverage of source genome
 - ▶ More than 30x average depth
- ▶ Two strategies:
 - ▶ De novo: assemble reads into a new sequence
 - ▶ Re-sequence: refine an existing reference sequence

De novo assembly

- **Genome Assembly** - Create new reference 'from scratch'
- Examine reads for overlapping sequence
- **Contig** - longer assembled sequence from short reads
- **Scaffold** - assembled contigs
- **Chromosome** - assembled scaffolds
- Assembly from short reads is hard

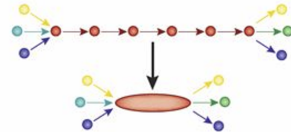
1. Fragment DNA and sequence



2. Find overlaps between reads

...AGCCTAGACCTACA**GGATGCGCGACACGT**
GGATGCGCGACACGTCGCATATCCGGT

3. Assemble overlaps into contigs



4. Assemble contigs into scaffolds

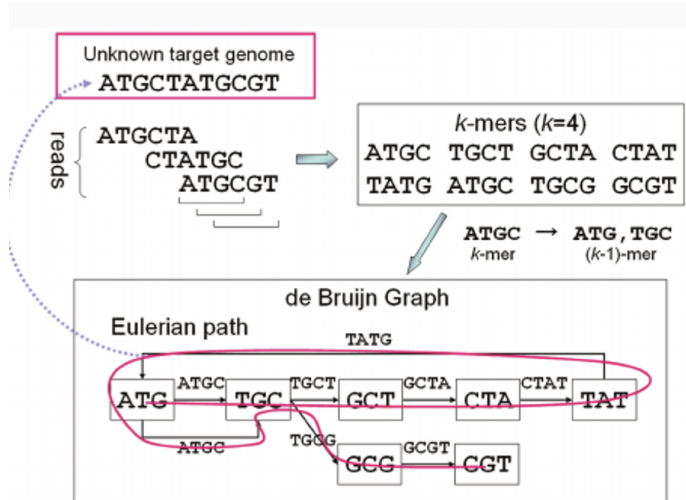


Michael Schatz, Cold Spring Harbor

De novo assembly: graph-based

- ▶ Greedy assembly creates linear sequences
- ▶ Vulnerable to finding local optima
- ▶ Graph representation considers all sequence content simultaneously
- ▶ Graph data structure can encode variability (e.g. insertions, SNPs)
- ▶ Computationally much more expensive
- ▶ de Bruijn Graphs and Overlap Layout Consensus

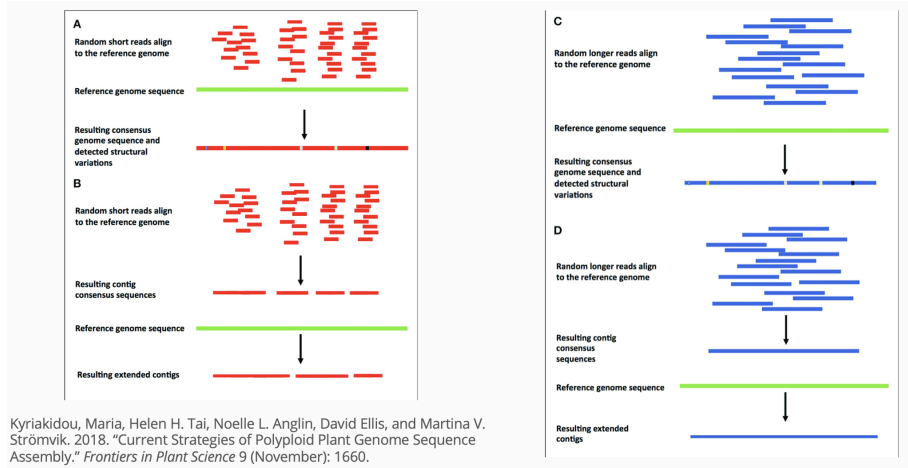
De novo assembly: graph-based



Reference guided genome assembly

- ▶ Refine existing reference with new sequence
- ▶ Can discover:
 - ▶ New structural variants
 - ▶ Novel insertions/alternate haplotypes or scaffolds
 - ▶ Polymorphisms
- ▶ Faster, easier than de novo assembly
- ▶ More sensitive to existing biases in reference

Reference guided genome assembly

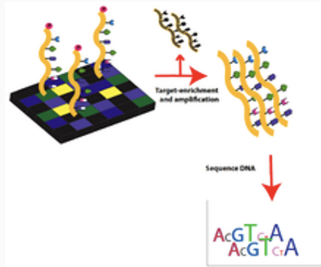
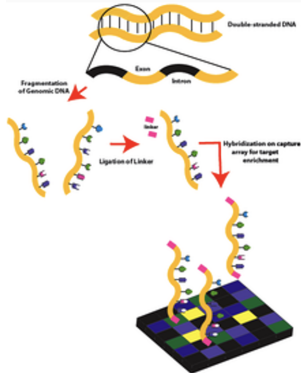


Whole Exome Sequencing

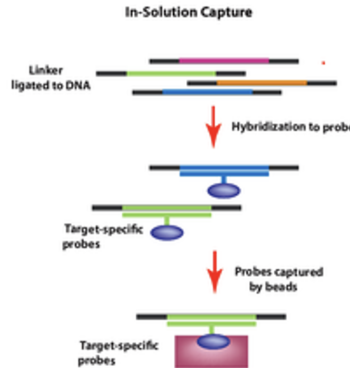
- ▶ **W**hole **E**xome **S**equencing (WES)
- ▶ Exons 1%-2% of human genome sequence
- ▶ Pre-select reads that map to exons
- ▶ Sequence to much greater depth than WGS
- ▶ Identify coding variants

Exome Sequence Selection

Array-based capture



In-solution

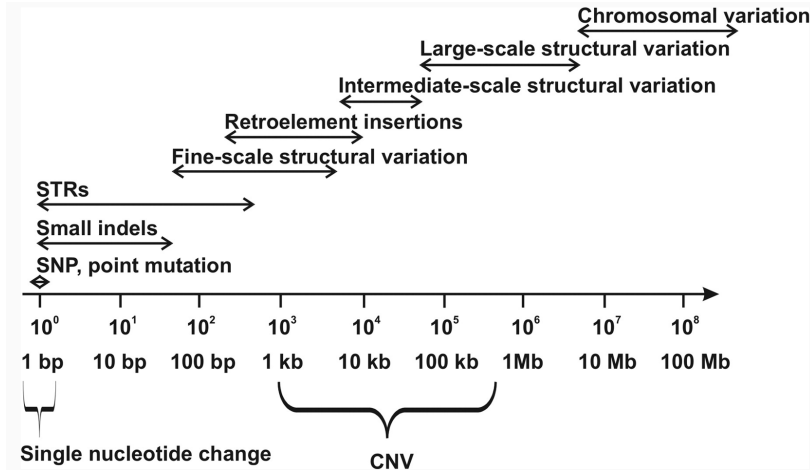


SNP and Variant Calling

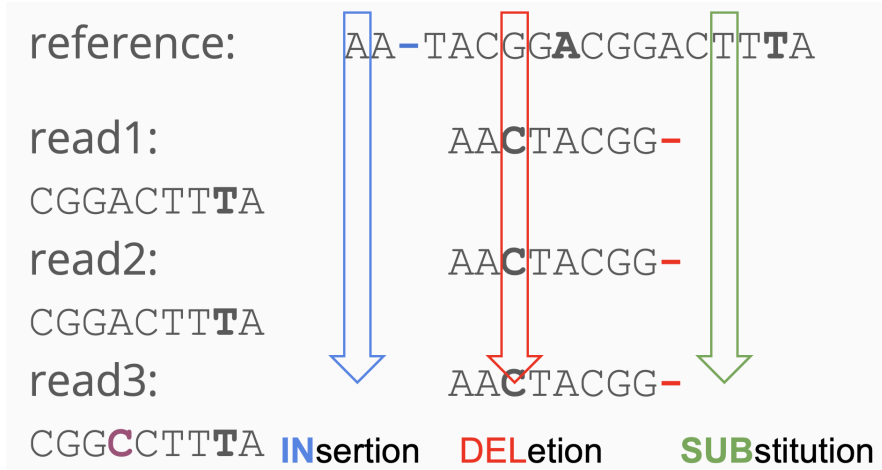
Genomic Variants

- ▶ Individual genomes from same species vary
- ▶ WGS/WES compared with reference can identify differences
- ▶ **Variant**: sequence that varies within species
- ▶ Two general types:
 - ▶ Small: <50 bp, single nucleotide polymorphisms (SNPs), indels
 - ▶ Large: >50 bp, copy number variations, duplications, deletions, translocations, inversions

Genomic Variants



Point mutations



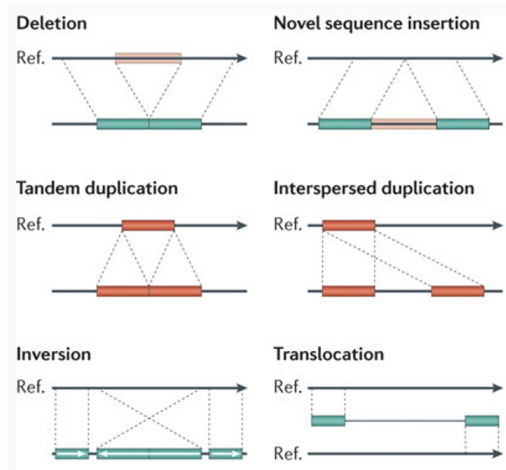
Single Nucleotide Polymorphisms (SNPs)

- ▶ Most commonly studied type of variant
- ▶ Types of single nucleotide alterations:
 - ▶ **Variant** (SNV) - any single mutation
 - ▶ **Polymorphism** (SNP) - SNV observed in significant frequency in population (e.g. $>1\%$)
- ▶ Typically base changes, e.g. A to C
- ▶ SNPs (usually) indicate shared ancestry
- ▶ May suggest disease mechanism

Structural variation

- ▶ Deletions - sequence missing
- ▶ Insertions:
 - ▶ Novel - new sequence added
 - ▶ Mobile-element - copied/moved from elsewhere in genome
- ▶ Duplications:
 - ▶ Tandem - consecutive
 - ▶ Interspersed - non-consecutive
- ▶ Inversion - segment is reversed
- ▶ Translocation - segment moves

Structural variation



Genotyping

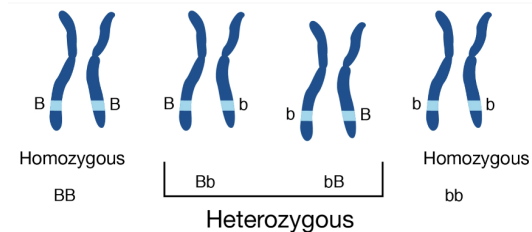
- ▶ **Genotype:** an individual's variant(s)
- ▶ **Phenotype:** an individual's physical form
- ▶ **De novo** variant calling/detection: given genomic reads and a reference, find all the variants
- ▶ **Genotyping:** examine individual for *a priori* variants at known locations
 - ▶ Can use arrays (i.e. SNP Chip) or WGS/WES

Human Genomic Variants

- ▶ Humans have diploid genome
 - ▶ i.e. 2 copies of each gene
- ▶ dbSNP - NCBI repository for SNPs
 - ▶ As of June 2021 (Build 155)
 - ▶ 3.3B submitted variants
 - ▶ 1B annotated human SNPs Every human has on average:
 - ▶ A variant every 1kb
 - ▶ 2-3 million SNPs

Human Genomic Terminology

- ▶ **Allele:** sequence containing a variant
- ▶ **Homozygous variant:** both same allele
 - ▶ i.e. either same variant or same as reference
- ▶ **Heterozygous variant:** different alleles
 - ▶ i.e. one is variant, one is reference/different variant



Human Genomic Terminology

For coding (exonic) variants:

- ▶ **Synonymous or sense:** variant does not change amino acid sequence
- ▶ **Non-synonymous or mis-sense:** variant causes amino acid change
- ▶ **Non-sense:** causes early termination of protein by introducing stop codon
- ▶ **Frameshift:** insertion or deletion causes complete recoding of downstream proteins

Genomic Variant Terminology

- ▶ **Germline:** Inherited from parents
 - ▶ e.g. blue eyes, familial disease risk
- ▶ **Somatic:** Acquired during life
 - ▶ e.g. tumor vs normal tissue
- ▶ **Allele frequency:** how common is a given variant in some population, e.g.:
 - ▶ 1% of human population
 - ▶ 30% within people with some disease

dbSNP - NCBI SNP Database

dbSNP Short Genetic Variations

Search for terms

Search

Examples: rs268, BRCA1 and more

Advanced search



Welcome to the Reference SNP (rs) Report

All alleles are reported in the [Forward orientation](#). Click on the [Variant Details](#) tab for details on Genomic Placement, Gene, and Amino Acid changes. HGVS names are in the [HGVS](#) tab.

Reference SNP (rs) Report

[Switch to classic site](#)

[Download](#)



rs429358

Current Build 155
Released April 9, 2021

Organism *Homo sapiens*

Clinical Significance Reported in [ClinVar](#)

Position chr19:44908684 (GRCh38.p13) [?](#)

Gene : Consequence APOE : Missense Variant

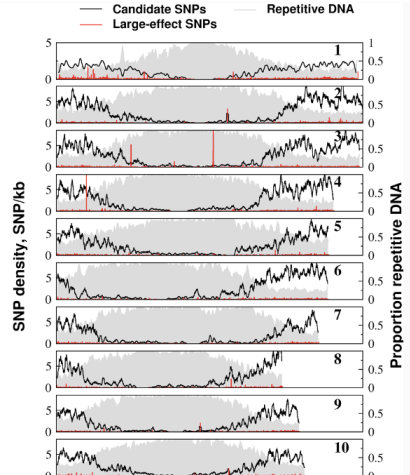
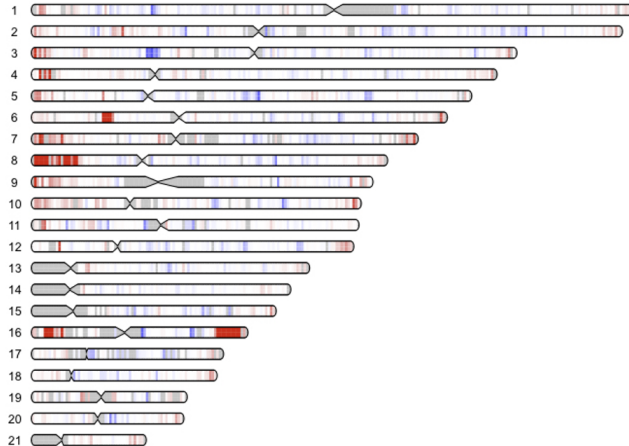
Alleles T>C

Publications [416 citations](#)

Variants Smaller Than A Read

- ▶ Finding SNPs, indels almost a solved problem
- ▶ SNPs called are 95% accurate
 - ▶ i.e. with sufficient coverage
- ▶ Structural variants cause false positives
- ▶ Duplications, somatic mutations may cause 3 or more alleles to be observed

SNP and indel density is non-random



Variants Larger Than A Read

- ▶ **Structural Variation (SV)**

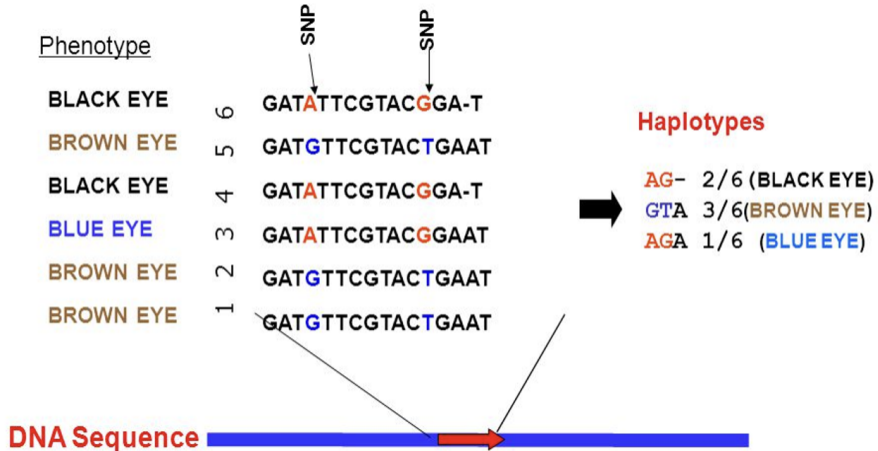
- ▶ Two types:

- ▶ Balanced - Do not change amount of DNA
- ▶ Copy Number Variants (CNV) - Change amount of DNA Scales:
 - ▶ Mini (hundreds of basepairs)
 - ▶ Macro (visible by a microscope) variants
 - ▶ Much harder to find (especially balanced) Non-random: SV 'hotspots'

Haplotyping

- ▶ DNA recombines in large blocks
- ▶ SNPs in a block move around together
- ▶ Looking at the common SNPs in a block, reveals the ancestry information
- ▶ **Linkage Disequilibrium (LD)**: adjacent SNPs co-occur more often than expected
 - ▶ i.e. 2 SNPs are in LD with each other

Haplotyping



Methods for SNP Calling

1. GATK (Genome Analysis Toolkit):

- ▶ Developed by the Broad Institute, GATK is a widely used toolkit for variant discovery in high-throughput sequencing data.
- ▶ It employs a best practices pipeline for variant calling, including base quality score recalibration, indel realignment, and variant quality score recalibration.
- ▶ GATK offers various tools such as HaplotypeCaller and UnifiedGenotyper for variant calling from both exome and whole genome sequencing data.

2. Mutect2:

- ▶ Mutect2 is part of the GATK toolkit and is specifically designed for somatic mutation calling, particularly in cancer genomes.
- ▶ It utilizes a probabilistic model to differentiate true somatic mutations from sequencing artifacts.
- ▶ Mutect2 can be applied to both exome and whole genome sequencing data to identify somatic variants.

Methods for SNP Calling

3. Samtools:

- ▶ Samtools is a suite of programs for interacting with high-throughput sequencing data in the SAM/BAM format.
- ▶ It includes the mpileup command for generating pileup data, and bcftools for variant calling from the pileup data.
- ▶ Samtools is efficient and widely used for variant calling in both exome and whole genome sequencing datasets.

4. FreeBayes:

- ▶ FreeBayes is a Bayesian genetic variant detector designed to detect SNPs, indels, and complex polymorphisms in high-throughput sequencing data.
- ▶ It utilizes a haplotype-based approach to increase sensitivity and specificity, particularly in the context of population-scale sequencing data.
- ▶ FreeBayes can be used for variant calling in both exome and whole genome sequencing studies.

Methods for SNP Calling

5. VarScan:

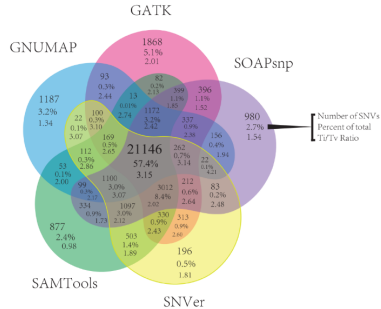
- ▶ VarScan is a platform-independent tool for variant detection in massively parallel sequencing data.
- ▶ It is optimized for calling germline variants, somatic mutations, and copy number alterations.
- ▶ VarScan supports both exome and whole genome sequencing data and provides a range of options for variant calling and filtering.

6. DeepVariant:

- ▶ DeepVariant is an end-to-end deep learning-based variant caller developed by Google.
- ▶ It employs a convolutional neural network (CNN) architecture to call SNPs and indels with high accuracy.
- ▶ DeepVariant is particularly effective in identifying complex variants and can be applied to both exome and whole genome sequencing data.

Inconsistencies Among SNP Callers

Low concordance of variant-calling pipelines (O'Rawe, *Genome Med*, 2013)



BWA Revisted (OnDemand Amarel Desktop)

```
# load bwa
module load bwa

# Index the genome:
bwa index genomefile.fa

# Align the reads and include a `read group' header, e.g.:
bwa mem \
  -R "@RG\tID:group1\tSM:sam1\tPL:illumina\tLB:lib1\tPU:unit1" \
  genomefile.fa myfastqfile.fq > bwaalign.sam
```

SAMtools Example

Multiple sample SNP calling:

```
## Load samtools
```

```
module load samtools
```

```
# Convert to BAM, sort, and index:
```

```
samtools view -bS myalignments.sam > myalignments.bam
```

```
samtools sort myalignments.bam -o myalignments.sorted.bam
```

```
samtools index myalignments.sorted.bam
```

```
## mpileup
```

```
samtools mpileup -f genomefile.fa \  
    myalignments.sorted.bam > myalignments.pileup
```

GATK Example

```
module load samtools
# Index reference
samtools faidx chrX_5MB.fa

module load gatk
# make sequence dictionary
gatk CreateSequenceDictionary -R chrX_5MB.fa
# make .vcf file!
gatk HaplotypeCaller \
    -R chrX_5MB.fa -I proband_short_bwa.sorted.bam \
    -O proband_short.vcf
```

VCF files

8 fixed columns: #CHROM, POS, ID, REF, ALT, QUAL, FILTER, INFO, FORMAT
Additional columns, one for each sample, with sample ID

(a) VCF example

```
##fileformat=VCFv4.1
##fileDate=20110413
##source=VCFtools
##reference=file:///refs/human_NCBI36.fasta
##contig=<ID=1,length=249250621,md5=1b22b98cdeb4a9304cb5d48026a85128,species="Homo Sapiens">
##contig=<ID=X,length=155270560,md5=7e0e2e580297b7764e31dbc80c2540dd,species="Homo Sapiens">
##INFO=<ID=AA,Number=1,Type=String,Description="Ancestral Allele">
##INFO=<ID=H2,Number=0,Type=Flag,Description="HapMap2 membership">
##FORMAT=<ID=GT,Number=1,Type=String,Description="Genotype">
##FORMAT=<ID=GQ,Number=1,Type=Integer,Description="Genotype Quality">
##FORMAT=<ID=DP,Number=1,Type=Integer,Description="Read Depth">
##ALT=<ID=DEL,Description="Deletion">
##INFO=<ID=SVTYPE,Number=1,Type=String,Description="Type of structural variant">
##INFO=<ID=END,Number=1,Type=Integer,Description="End position of the variant">
```

#CHROM	POS	ID	REF	ALT	QUAL	FILTER	INFO	FORMAT	SAMPLE1	SAMPLE2
1	1	.	ACG	A,AT	40	PASS	.	GT:DP	1/1:13	2/2:29
1	2	.	C	T,CT	.	PASS	H2;AA=T	GT	0 1	2/2
1	5	rs12	A	G	67	PASS	.	GT:DP	1 0:16	2/2:20
X	100	.	T		.	PASS	SVTYPE=DEL;END=200	GT:GQ:DP	1:12:	0/0:20:36

Complete Variant Calling Pipeline (Outdated!)

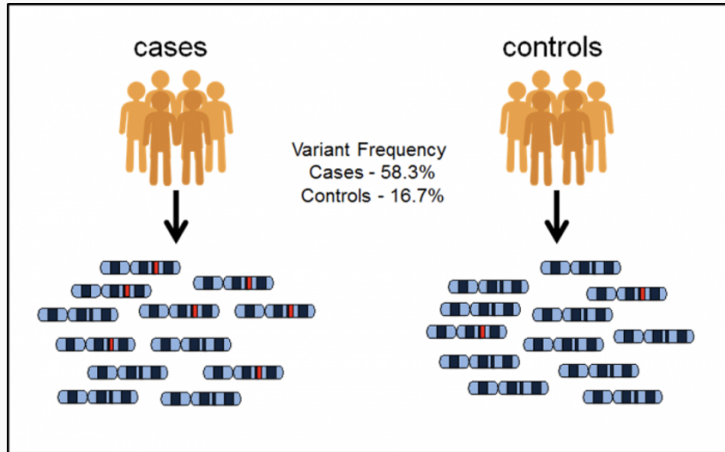
Our analysis pipeline consisted of the following:

- ▶ Align the FASTQ files to genome
- ▶ Convert SAM file to BAM, add read group info
- ▶ Filter the reads based on quality (BAMTools)
- ▶ Samtools to sort and index, and use Picard to mark duplicates
- ▶ GATK calibration, realignment, variant calling (HaplotypeCaller, Mutect2)
- ▶ Filter the called variants (GATK filtersnp and filterindels).
- ▶ Annotation of SNPs (snpEff, condel)
- ▶ Filter by frequency (thousand genomes, TCGA, etc.)
- ▶ Downstream analysis (rare variants, pedigree, pathway level, etc)

Genome Wide Association Studies (GWAS)

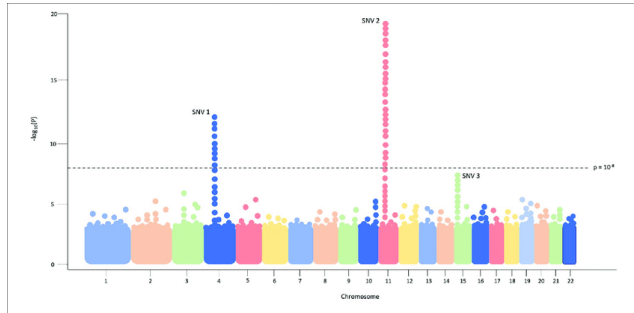
- ▶ Which SNPs are associated with a variable of interest?
 - ▶ e.g. disease, height Does the frequency of any SNP differ between groups? Associated SNPs have:
- ▶ Effect size - e.g. amount of increased risk
- ▶ p-value - precision of effect
- ▶ **Risk allele**: allele associated with increased or decreased probability of having a disease

Genome Wide Association Studies (GWAS)



Manhattan Plot

- ▶ In a **Manhattan plot**, each dot represents a genetic marker (SNP) on the x-axis, and its $-\log_{10}(\text{p-value})$ on the y-axis.
- ▶ The horizontal lines represent the significance threshold.
- ▶ Peaks represent regions where genetic variants are significantly associated with the trait or disease.



Genome-Wide Significance Threshold

- ▶ The genome-wide significance threshold is a threshold used to determine if an association between a genetic variant and a trait is statistically significant.
- ▶ It takes into account the multiple comparisons problem arising from testing thousands or millions of genetic markers across the genome.

Bonferroni Correction

- ▶ One method to establish the genome-wide significance threshold is the Bonferroni correction.
- ▶ The Bonferroni-corrected threshold is calculated by dividing the desired significance level (e.g., 0.05) by the number of tests performed.
- ▶ The most commonly accepted threshold is

$$p < 5 \times 10^{-8},$$

based on a Bonferroni correction for all independent common SNPs across the human genome

False Discovery Rate (FDR)

- ▶ Another method to control for multiple comparisons is the False Discovery Rate (FDR).
- ▶ FDR controls the proportion of false positives among all significant results.
- ▶ It is less conservative than the Bonferroni correction and allows for a higher number of false positives while still controlling the overall error rate.

Downstream Annotation and Analysis (Outdated!)

Downstream Annotation Tools (old list):

- ▶ snpEff (<http://snpeff.sourceforge.net/>)
- ▶ Condel (<http://bg.upf.edu/condel/home>)
- ▶ SIFT <http://sift.jcvi.org/>
- ▶ Polyphen 2 <http://genetics.bwh.harvard.edu/pph2/>
- ▶ <http://mutationassessor.org/>
- ▶ Ensembl variant effect predictor
(<http://www.ensembl.org/info/docs/variation/vep/index.html>)
- ▶ Thousand Genomes variant frequency (e.g. 1% threshold) and Exome Sequence Project variant frequency (e.g. 1%).

Session info

```
sessionInfo()
```

```
## R version 4.4.2 (2024-10-31)
## Platform: aarch64-apple-darwin20
## Running under: macOS Sequoia 15.5
##
## Matrix products: default
## BLAS:   /Library/Frameworks/R.framework/Versions/4.4-arm64/Resources/lib/libRblas.0.dylib
## LAPACK: /Library/Frameworks/R.framework/Versions/4.4-arm64/Resources/lib/libRlapack.dylib; LAPACK version 3.12.0
##
## locale:
## [1] en_US.UTF-8/en_US.UTF-8/en_US.UTF-8/C/en_US.UTF-8/en_US.UTF-8
##
## time zone: America/Denver
## tzcode source: internal
##
## attached base packages:
## [1] stats      graphics  grDevices utils      datasets  methods   base
##
## loaded via a namespace (and not attached):
## [1] compiler_4.4.2 fastmap_1.2.0 cli_3.6.5      tools_4.4.2
## [5] htmltools_0.5.8.1 rstudioapi_0.17.1 yaml_2.3.10    rmarkdown_2.29
## [9] knitr_1.50      xfun_0.52      digest_0.6.37  rlang_1.1.6
## [13] evaluate_1.0.3
```